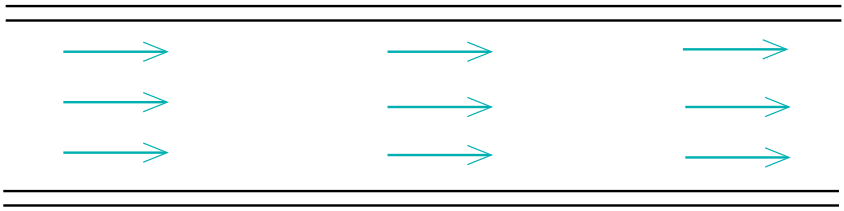
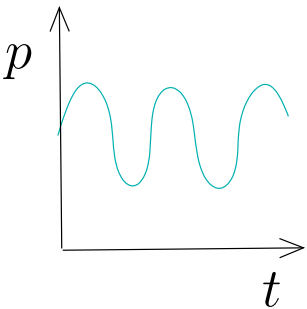
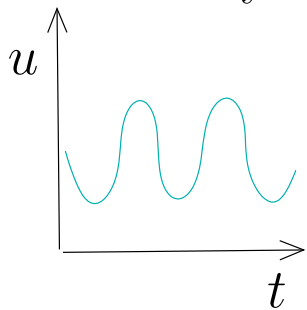


p, u spatially invariant

pressure $p(t)$



velocity $u(t)$